

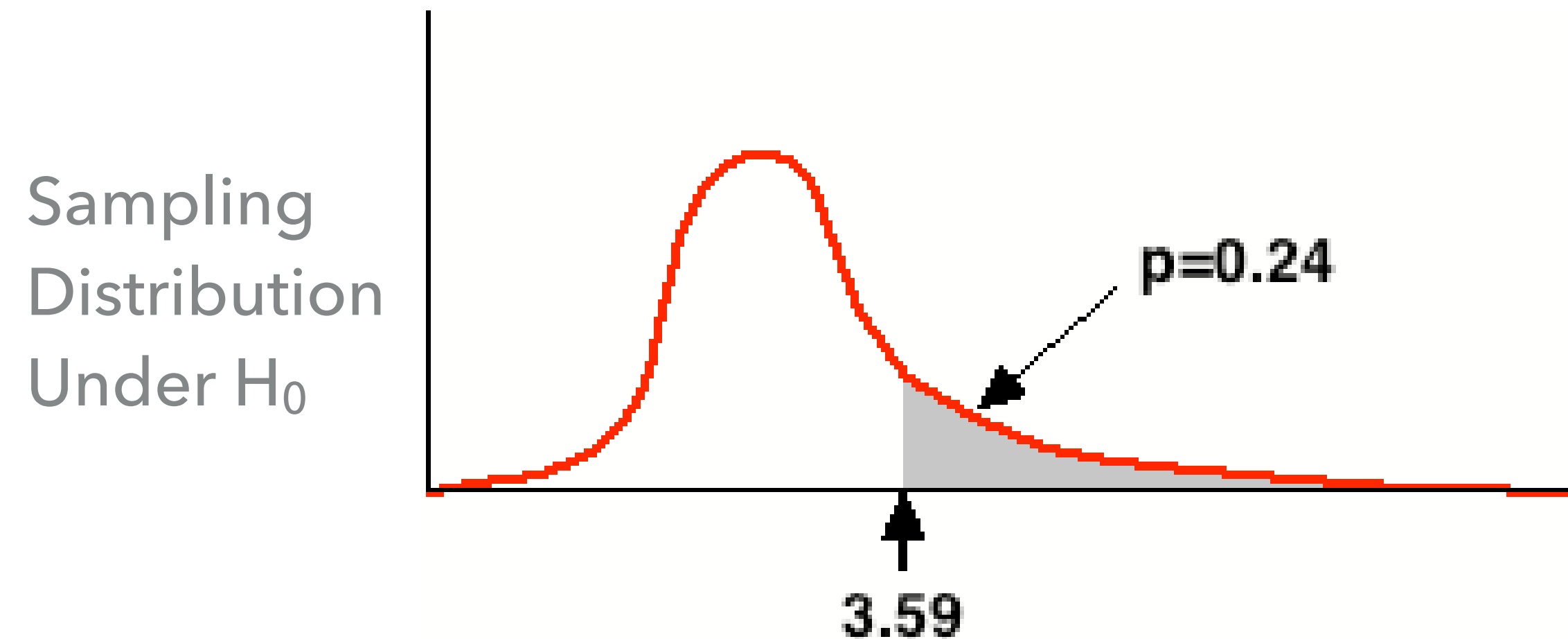
CS57300
PURDUE UNIVERSITY

DATA MINING

HYPOTHESIS TESTING

STATISTICAL SIGNIFICANCE

- ▶ A value of a statistic is **statistically significant** if it is unlikely to occur under the null hypothesis



**significance
level**

$$\alpha = p(\text{reject } H_0 | H_0 \text{ true}) = p(\text{type 1 error})$$

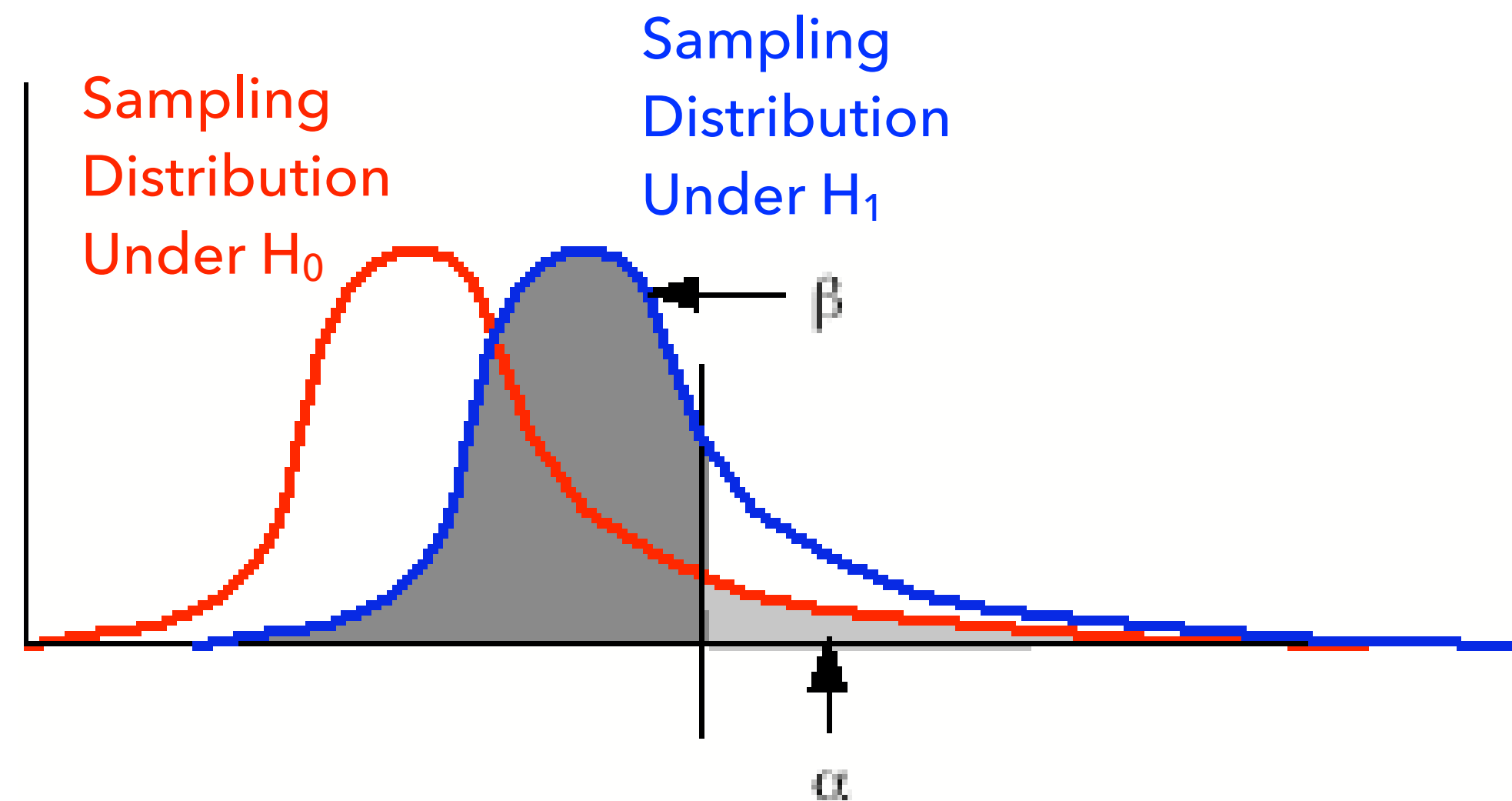
ERRORS

		Decision	
		Reject H_0	Don't reject H_0
Truth	H_0	<i>Type 1 error</i>	
	H_1		<i>Type 2 error</i>

- ▶ Type 1: null is rejected when it is true
 - ▶ E.g., conclude cancer drug increases life expectancy when in fact it doesn't
 - ▶ Generally considered to be most serious error
- ▶ Type 2: null is accepted when it is false
 - ▶ E.g., conclude that cancer drug does not increase life expectancy when in fact it does

STATISTICAL POWER

- ▶ Lack of statistical significance does not necessarily imply that H_0 is true
- ▶ Test could have low statistical power: $(1 - \beta)$ **portion of sampling distribution for alternative that is above threshold**



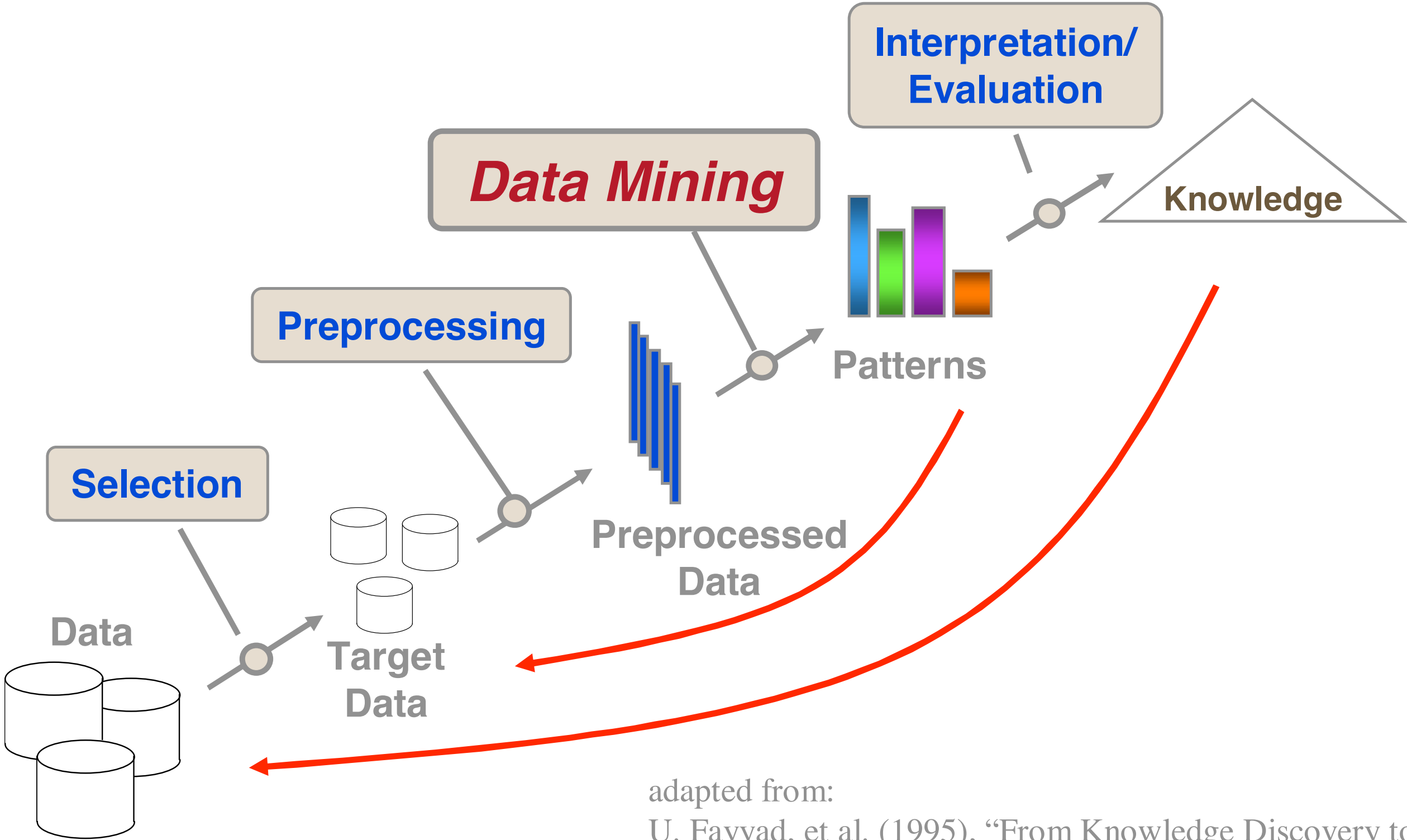
$$\beta = p(\text{accept } H_0 | H_0 \text{ false}) = p(\text{type 2 error})$$

HOW TO INCREASE POWER

- ▶ Increase sample size
- ▶ Decrease sample variability
 - ▶ Matching, sample selection, control for confounding variables, increase precision of measurements
- ▶ Increase effect size
 - ▶ More extreme experimental conditions, avoid ceiling/floor effects
- ▶ Increase alpha (e.g., from 0.05 to 0.10, but this increases type 1 errors)

ELEMENTS OF DATA MINING ALGORITHMS

DATA MINING PROCESS



adapted from:
U. Fayyad, et al. (1995), "From Knowledge Discovery to Data Mining: An Overview," Advances in Knowledge Discovery and Data Mining, U. Fayyad et al. (Eds.), AAAI/MIT Press

Data Mining, U. Fayyad et al. (Eds.), AAAI/MIT Press
Mining: An Overview,, Advances in Knowledge Discovery and
U. Fayyad et al. (Eds.), From Knowledge Discovery to Data

OVERVIEW

- ▶ Task specification
- ▶ Knowledge representation
- ▶ Learning technique
 - ▶ Search + scoring
- ▶ Prediction and/or interpretation

OVERVIEW

- ▶ **Task specification**
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TASK SPECIFICATION

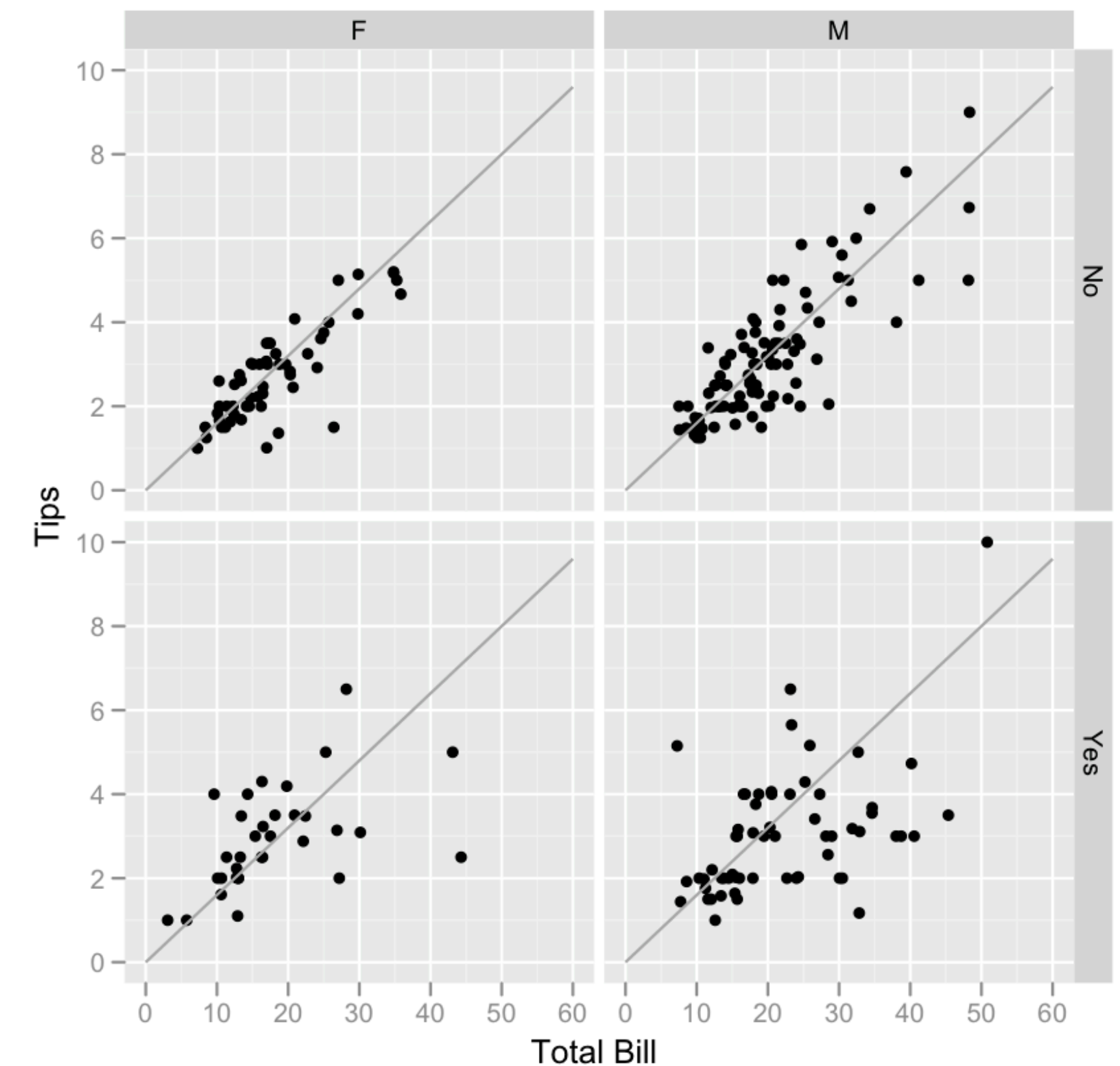
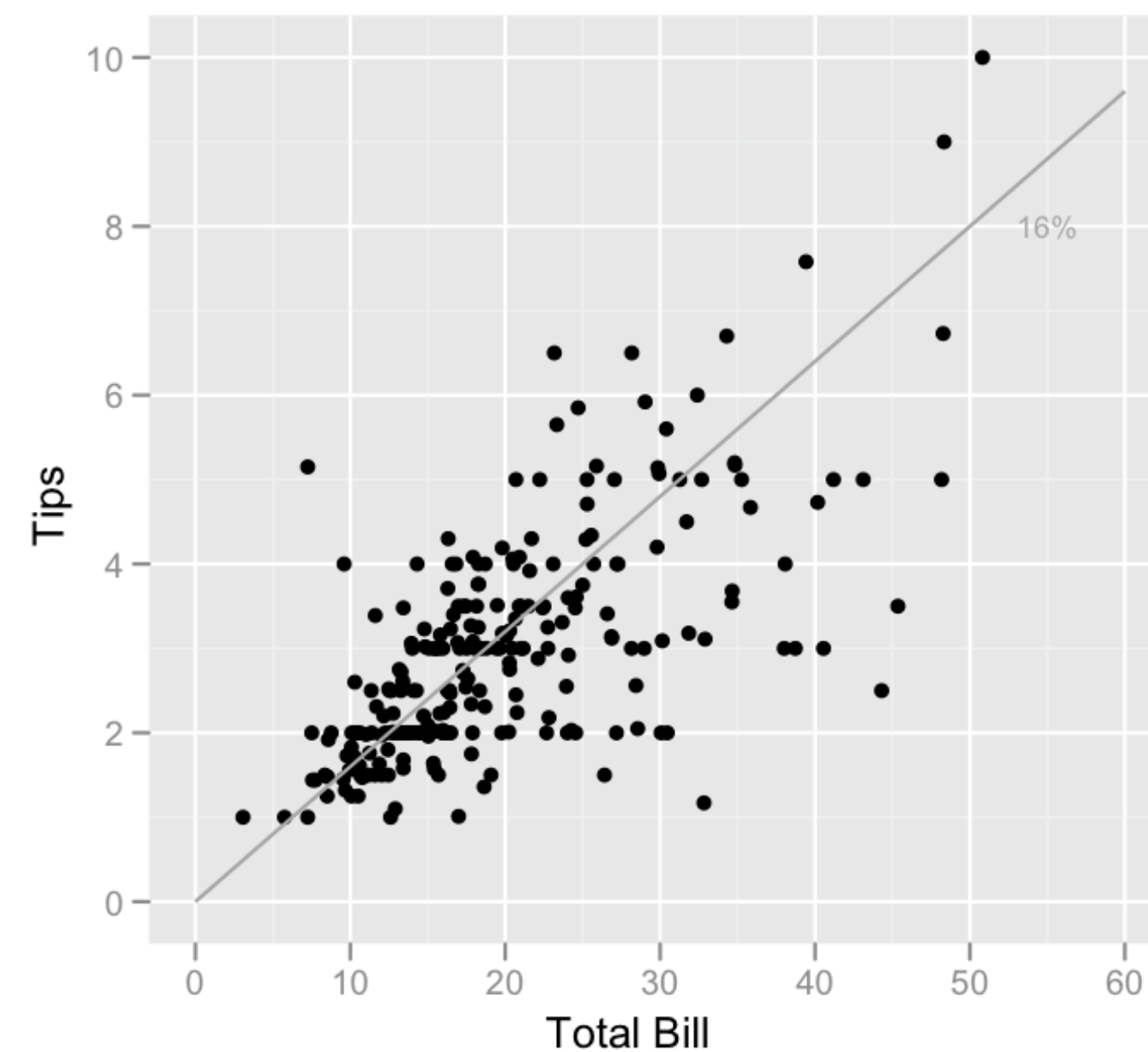
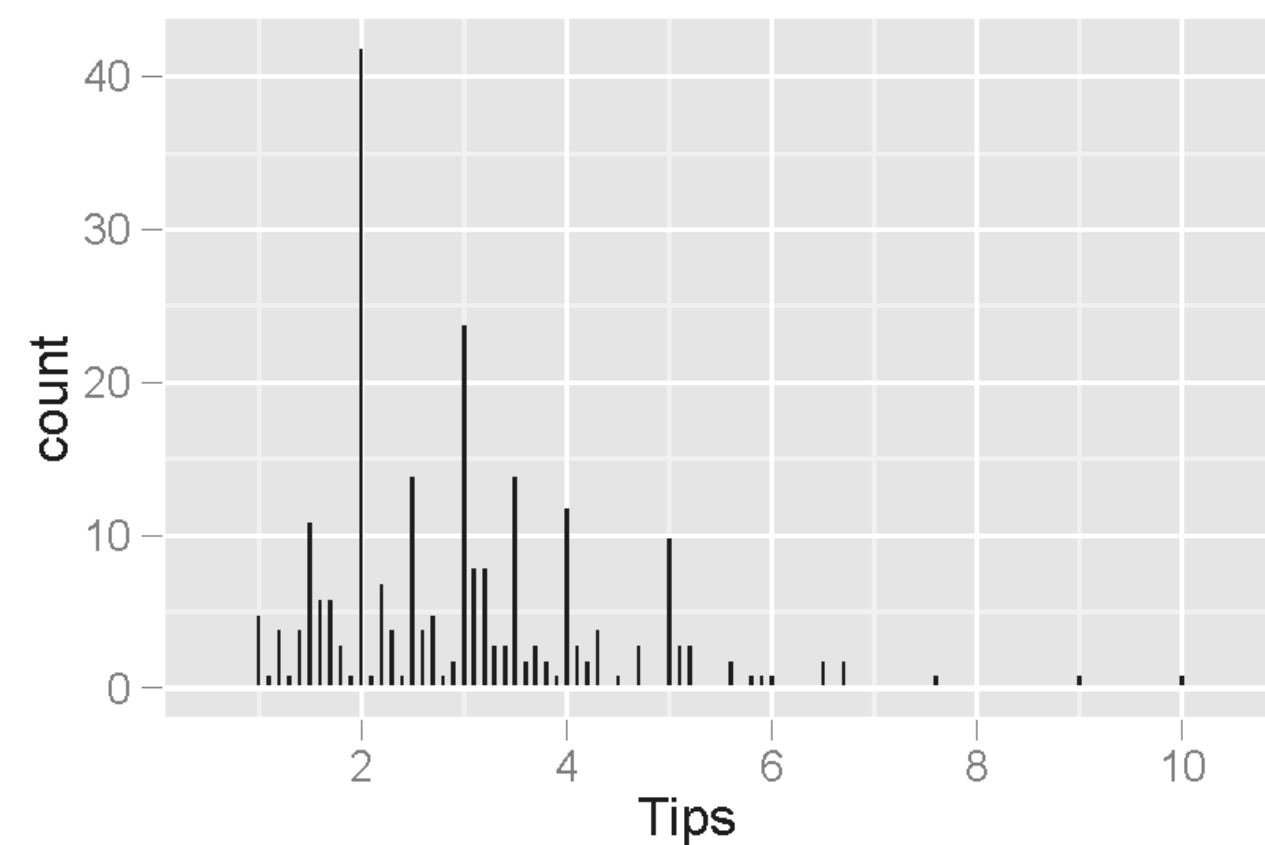
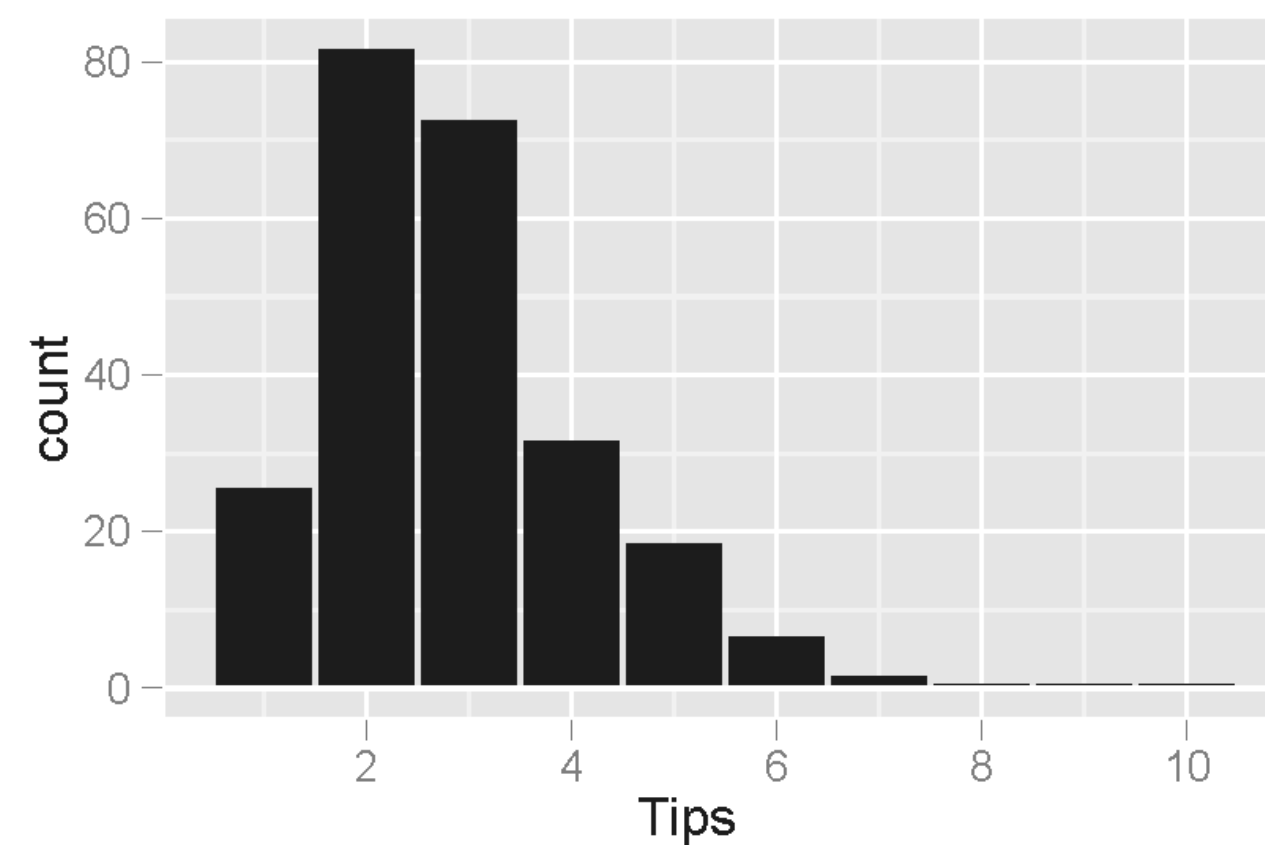
- ▶ Objective of the person who is analyzing the data
- ▶ Description of the characteristics of the analysis and desired result

EXPLORATORY DATA ANALYSIS

- ▶ Goal
 - ▶ Interact with data without clear objective
 - ▶ Summarize the main characteristics of the data
- ▶ Techniques
 - ▶ Mostly visualization

EXPLORATORY DATA ANALYSIS EXAMPLE

- What influences the amount of tip that a dining party will give to the waiter?



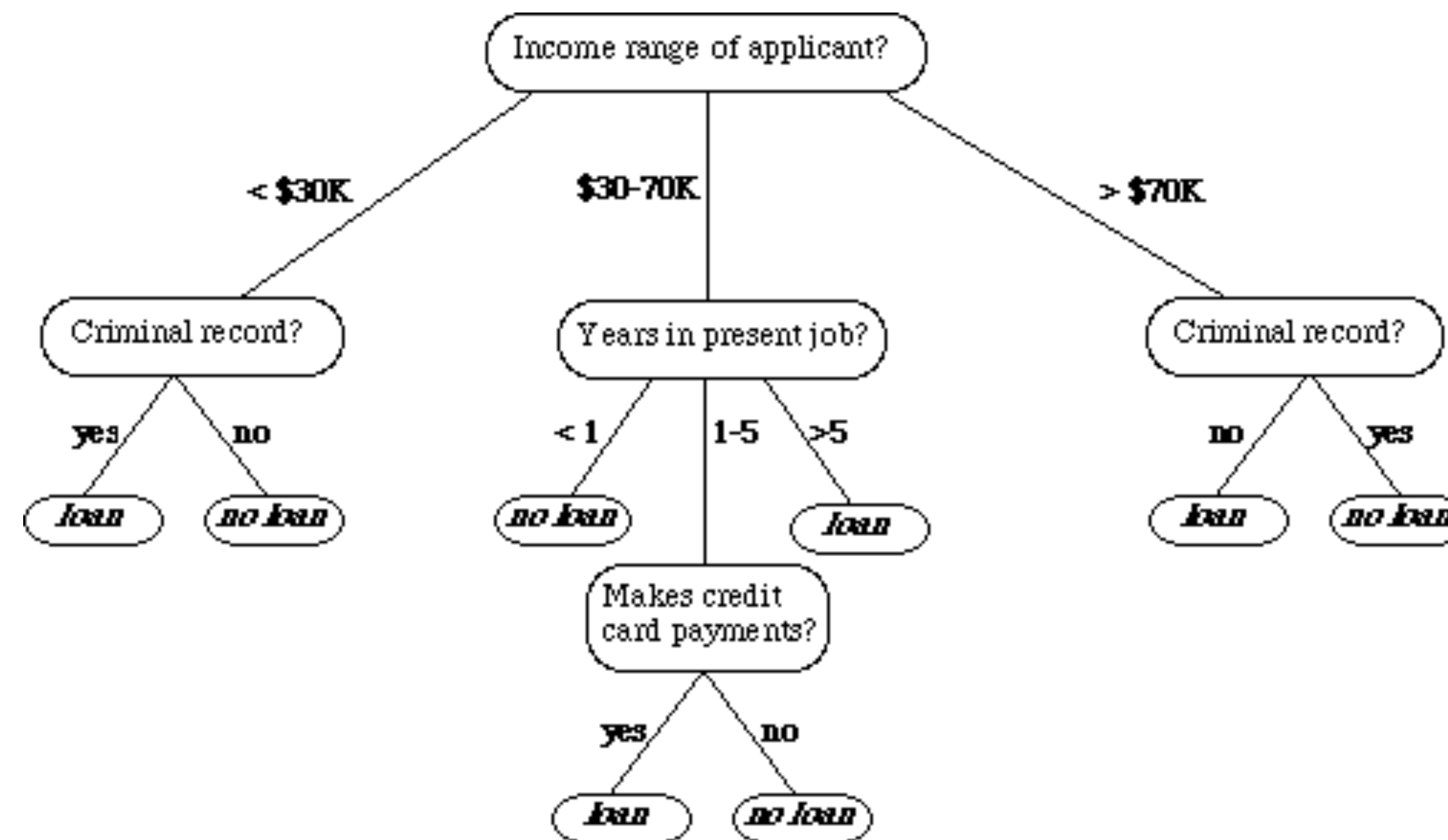
PREDICTIVE MODELING

- ▶ Goal
 - ▶ Learn model to predict the unknown value of a variable of interest given observed attribute values
- ▶ Techniques
 - ▶ Classification, regression

Also known as: **supervised** learning

PREDICTIVE MODELING EXAMPLE

- ▶ Zestimate: House sales price prediction!
- ▶ Predicting loan repayment (and thus decide whether to provide a loan)



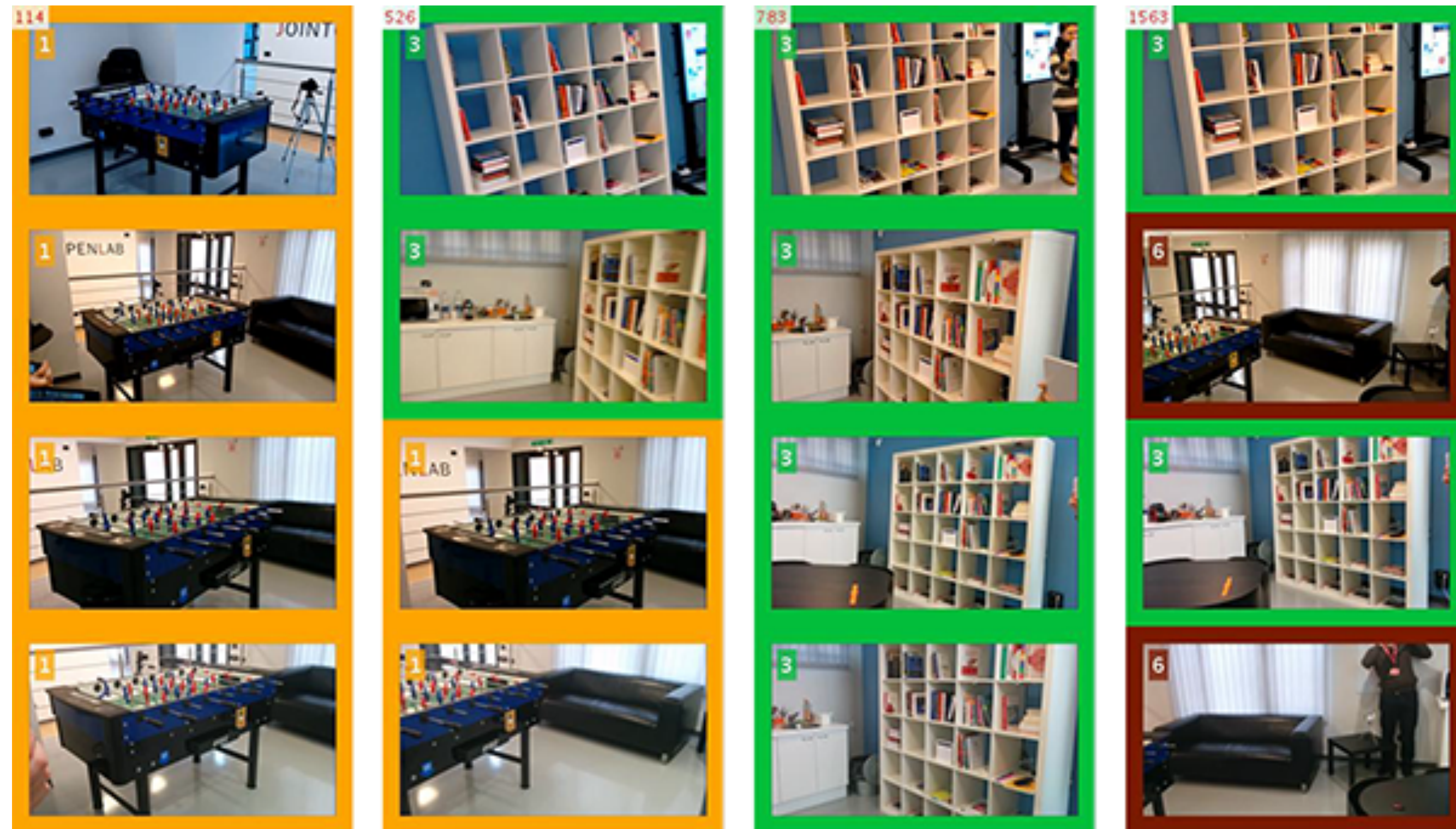
DESCRIPTIVE MODELING

- ▶ Goal
 - ▶ Summarize the data or the underlying generative process
- ▶ Techniques
 - ▶ Density estimation, cluster analysis and segmentation, probabilistic graphical model

Also known as: **unsupervised** learning

DESCRIPTIVE MODELING EXAMPLE

► Video/scene clustering



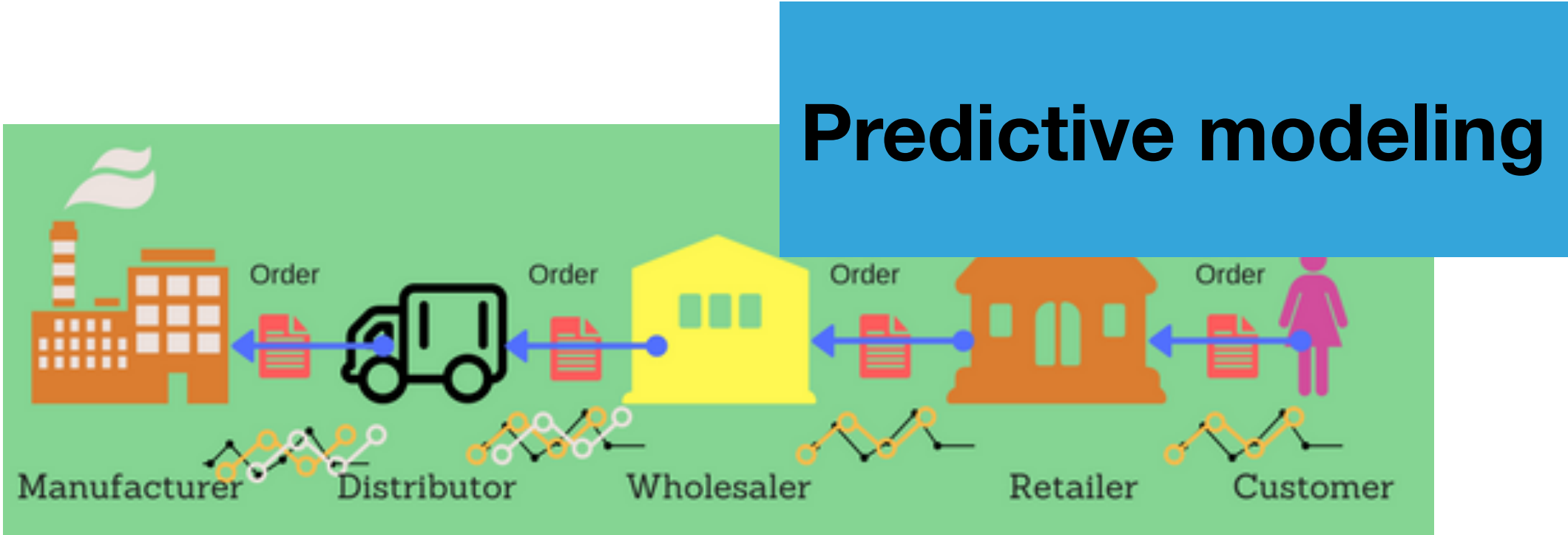
PATTERN DISCOVERY

- ▶ Goal
 - ▶ Detect patterns and rules that describe subsets of examples
- ▶ Techniques
 - ▶ Association rules, anomaly detection, etc.

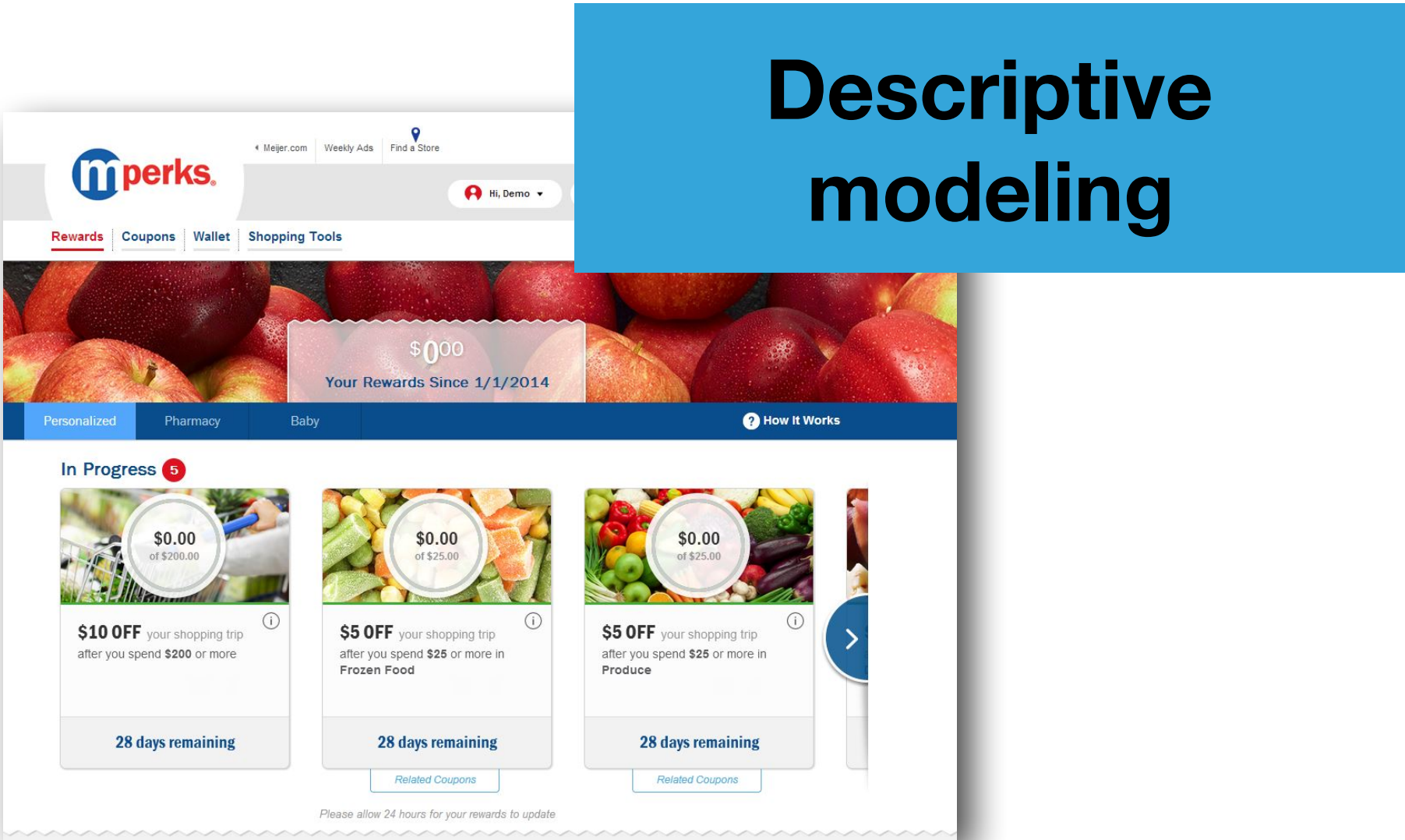
Model: global summary of a data set

Pattern: local to a subset of the data

WHAT TASKS ARE THEY?



Sales and inventory forecast



Customer segmentation



Pregnant custo



Beer & Dia